As drone technology keeps evolving at lightning speed, it introduces new and complex challenges to drone perimeter security, fundamentally altering the landscape for critical infrastructure protection.

So, what are the most pressing threats these increasingly sophisticated devices pose, and how can security measures evolve to counter them effectively?

In this article, we examine some of the perimeter security issues drones present and explore what it will take to safeguard sensitive areas in this rapidly advancing drone technology era.

What is perimeter security?

Perimeter security is the first line of defense for safeguarding physical spaces, assets, and infrastructure. It involves a strategic arrangement of barriers, surveillance, and detection systems to monitor and control access to a defined area, typically around critical infrastructure or restricted facilities.

Effective drone perimeter security relies on technologies like cameras, motion detectors, and sensors, as well as advanced tools like radar and radio frequency systems to detect unauthorized access or breaches. In today's environment, with the rise of increasingly

A new reality in perimeter security

As unmanned aerial systems (UAS) have risen in popularity, so has the frequency of drone-related incidents. The disruptive potential of drones became very clear in 2018 when Gatwick Airport faced an unprecedented shutdown due to a drone breaching its perimeter security.

The Gatwick incident showcased a new reality: drones represent a potent threat capable of causing widespread damage and disruption.

As drone technology rapidly evolves, security concerns for critical infrastructure have only intensified. Traditional security measures, designed to protect against physical and cyber threats, increasingly appear vulnerable to airborne threats that can circumvent traditional barriers altogether.

Benefits of using drones and automation in perimeter security

Using drones and automation in perimeter security offers a range of advantages, enhancing both efficiency and effectiveness. Using advanced technology, drones autonomously patrol designated areas, detecting and responding to potential security threats. Drones can rapidly cover large areas, providing real-time surveillance and monitoring of hard-to-reach locations, while automated systems enable continuous, 24/7 observation without the need for constant human oversight.

Equipped with advanced sensors and AI, perimeter security drones can detect and track intrusions with precision, reducing response times and improving situational awareness. This allows security teams to anticipate potential threats proactively, making perimeter security more adaptive and responsive to evolving risks.

6 of the challenges drones bring in terms of perimeter security

1. The current state of drone sophistication

able to avoid radar detection, deploy counter-detection measures, and more – making closing the window of vulnerability a significant challenge with no single solution.

2. The speed of evolution

Rapid domain evolution poses another challenge to perimeter security. Within such a fast-paced risk environment, security teams may often feel that they're at a disadvantage when ensuring that they confidently deploy C-UAS solutions.

Searching for a solution to the latest development can often involve teams seeking best-in-class sensor types from many third-party vendors. This can introduce its own problem: with several isolated sensors, how can teams access a clear operating picture to support rapid action? **Read more about this in our blog.**

3. Rapid response times are now vital

When drones cross protected perimeters – either through an unintended breach or for malicious purposes - every second matters. When drones are capable of causing significant damage or **scoping targets prior to a theft**, closing the window of vulnerability continues to remain a high priority.

However, many commercial drones are now able to reach incredibly high top speeds (**currently meeting an average of 50mph**). They can also bypass secure perimeters and enter secure airspace exceptionally quickly. This presents security teams with the task of responding to new risks in essentially real-time.

4. Intelligent onboard systems disrupt law enforcement

With onboard intelligence equipping drones with the benefits of AI, it is no longer necessary for human pilots to control drones. For law enforcement teams, this presents a new challenge – how to identify the drone's owner if there's no location data available to be intercepted.

This is an ongoing concern, and one that we're actively involved in tackling through our close collaboration with UK law enforcement, defence research, and other sectors.

5. Drones can cause significant harm

Drones can cause significant harm to people, property, or wider environments, such as when entering the turbines of passenger jet engines or falling from great heights into

Drones are also capable of presenting other forms of unique harm to security. Their use for **smuggling illegal contraband** into prisons has already been flagged as a serious risk, while new abilities for deploying payloads through a variety of means, **including chemical dispersions**, are a constant battle for security teams.

6. Forming responses is a complex procedure

Even when drones are detected, forming an appropriate response can be tricky. Currently, it is illegal for civilians to intervene in drone operations, regardless of if they are breaching secure airspace.

Forming an appropriate response to drone incursions is instead the sole responsibility of law enforcement. In the UK, a core priority for drone response is to locate the pilot responsible, and provide words of caution against future action, or make arrests in severe cases.

However, **as drone perimeter responses are set to become more complex and high-risk**, determining the safest course of action will be both more critical, and require a greater level of authority.

Providing security solutions to create safer spaces

At Operational Solutions, we are committed to equipping security teams with world-leading situational awareness and intelligence tools to enhance safety and readiness in the face of drone threats.

Our advanced drone detection systems and aerial surveillance capabilities empower security operations with real-time visibility, allowing teams to proactively address and close the window of vulnerability. Our solutions include:

- **Counter-drone systems** – Offering modular, scalable solutions that adapt to any facility's unique needs, enhancing detection and mitigation.
- **Intelligent ground security systems** – Using Perimeter Intrusion Detection Systems (PIDS) and Wide Area Detection Systems (WADS) for heightened perimeter protection.
- **Counter-drone consultancy** – Providing customised consultancy that enables a comprehensive understanding of situational intelligence and defence strategies.